

Database Benchmark Report

Automatically generated benchmark report

Date: 2026-07-02 13:49:45

Operating System: Linux

Python: 3.12.3

Databases: H2, HSQLDB

Benchmarks: 8

Measured Runs: 20

Warm-up Runs: 5

Benchmark Summary

Number of databases: **2**

Benchmark categories: **8**

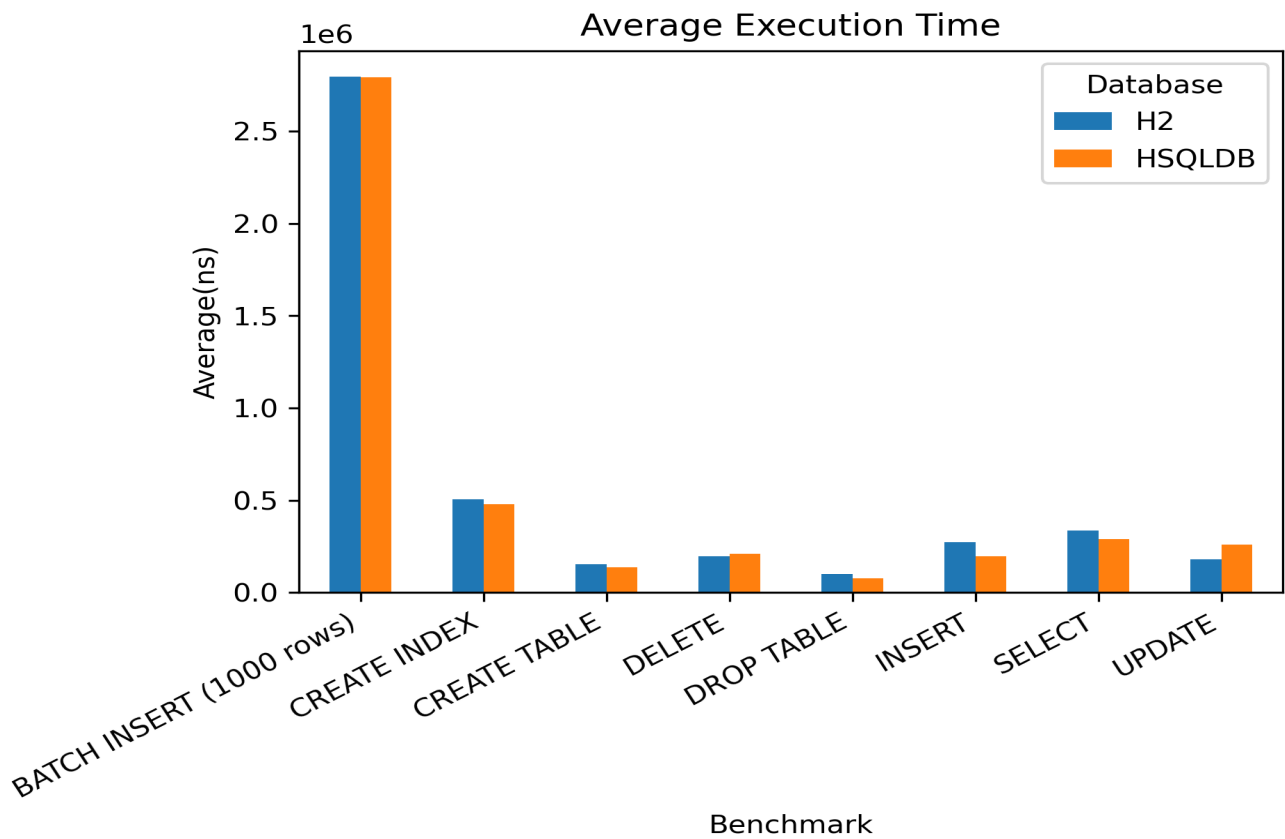
Benchmark results: **16**

This report summarizes execution-time measurements for embedded relational database management systems. All measurements were performed after JVM warm-up iterations and repeated multiple times to improve statistical reliability.

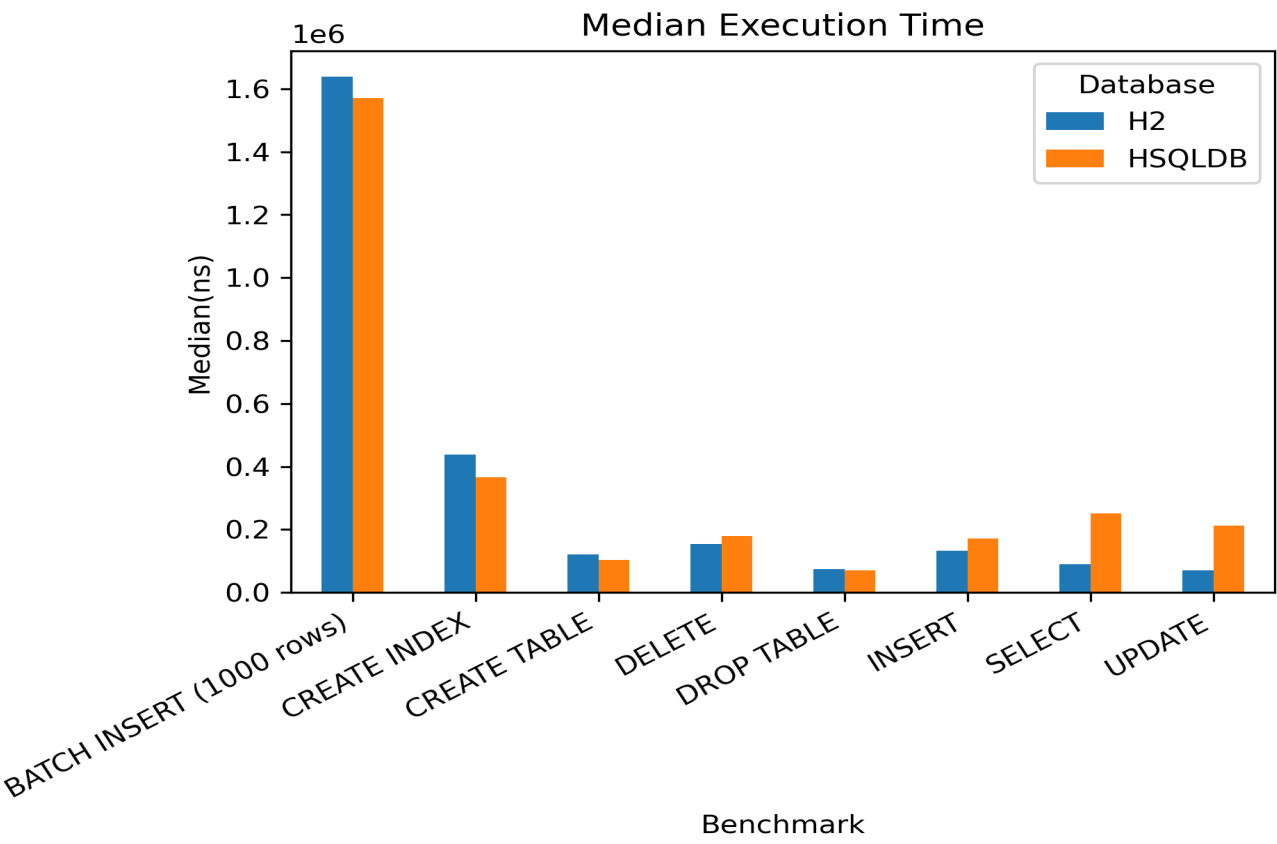
Summary Statistics

Database	Benchmark	Average	Median	Std Dev	Minimum	Maximum
H2	CREATE TABLE	151484	119937	142656	102384	3627836
H2	INSERT	274363	132346	464718	54072	3810743
H2	BATCH INSERT (1000 rows)	2796589	1638637	3317500	1320018	30317762
H2	SELECT	335594	88933	680044	56617	7500947
H2	UPDATE	179945	69336	264479	57639	3630501
H2	DELETE	196740	153877	219585	50055	2439086
H2	DROP TABLE	98736	74827	162954	64863	2584291
H2	CREATE INDEX	506012	437243	733022	354460	19547609
HSQLDB	CREATE TABLE	136250	102869	177749	60244	3297711
HSQLDB	INSERT	196635	171710	218356	132170	9317826
HSQLDB	BATCH INSERT (1000 rows)	2792928	1569936	15082348	967461	585053636
HSQLDB	SELECT	287690	250955	228301	172566	3195637
HSQLDB	UPDATE	259176	212668	148273	147269	2610050
HSQLDB	DELETE	210474	178278	124962	134866	3242957
HSQLDB	DROP TABLE	76289	69432	25681	51718	271835
HSQLDB	CREATE INDEX	478587	366468	619465	271214	12742889

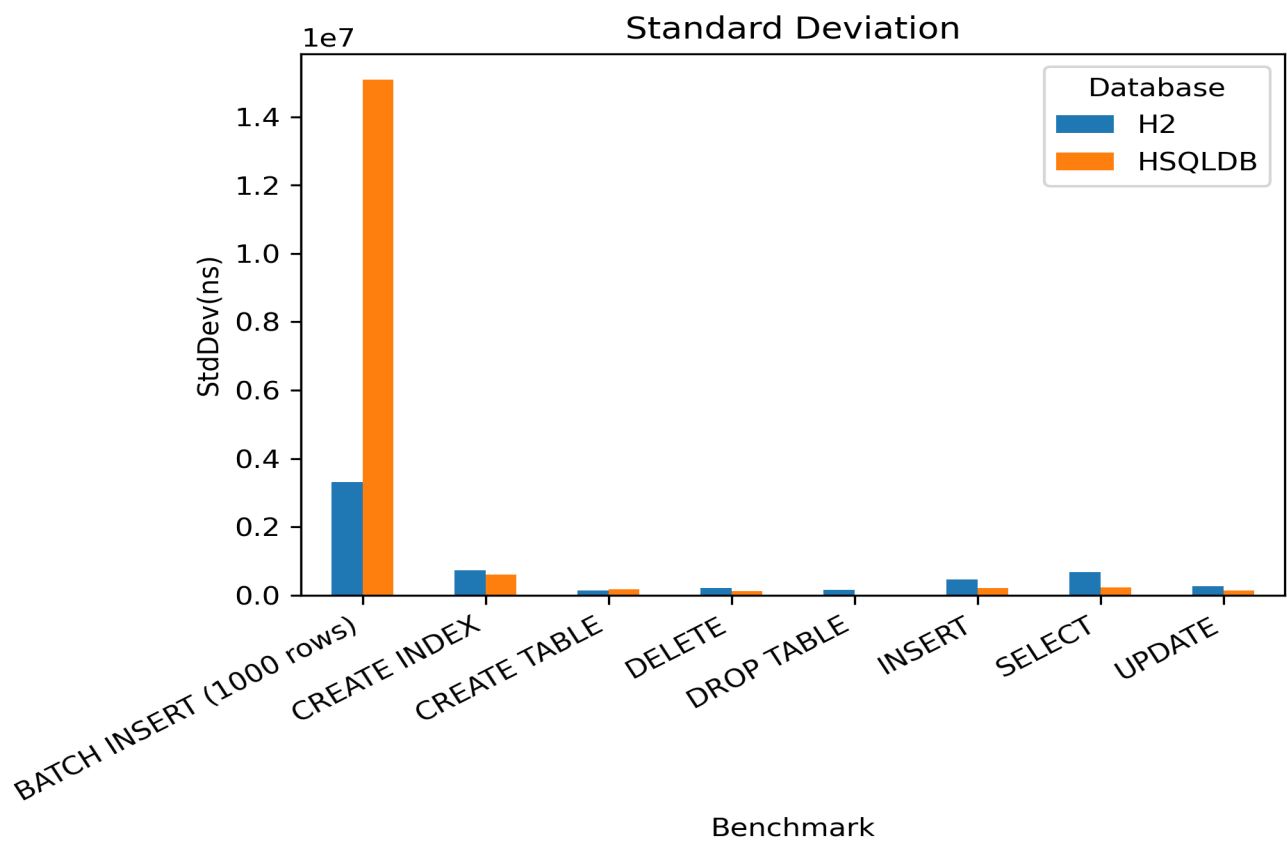
Average Execution Time



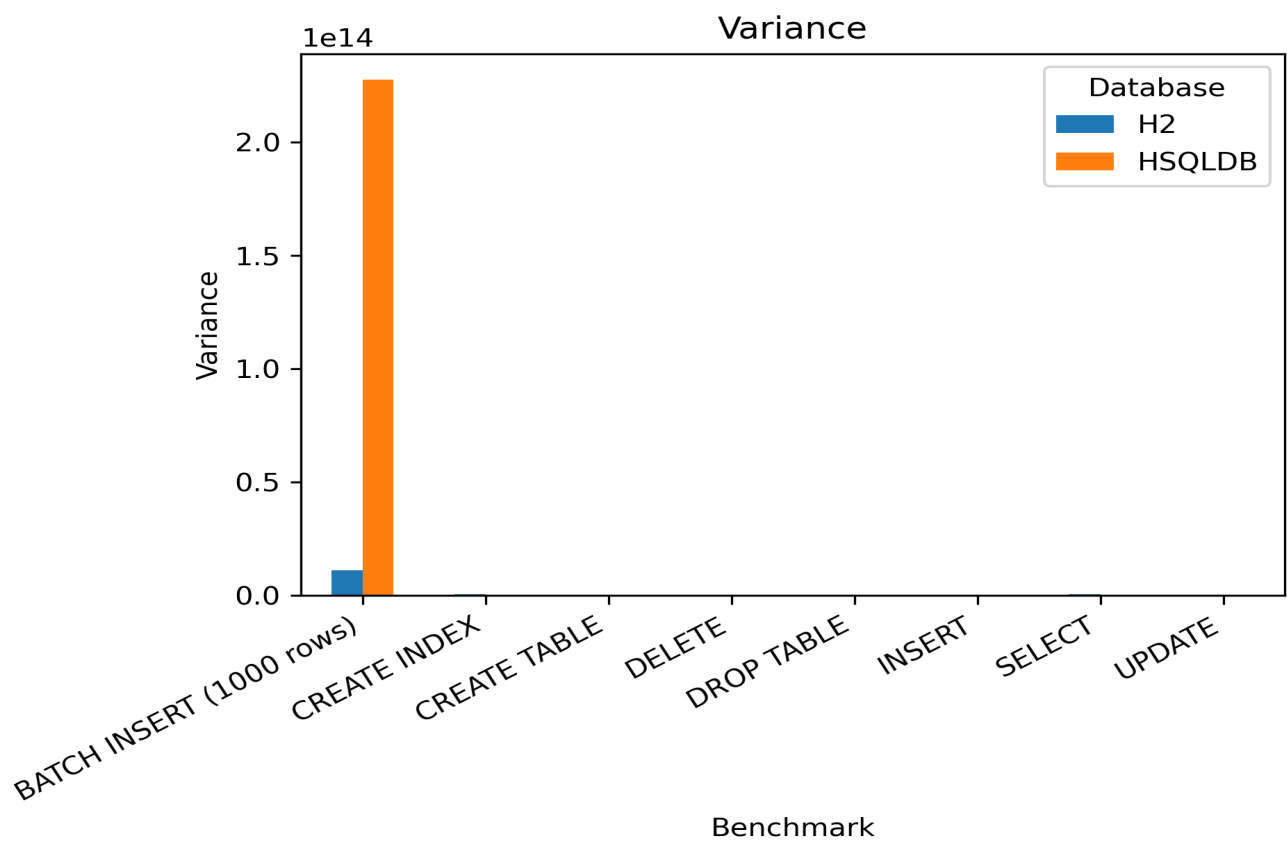
Median Execution Time



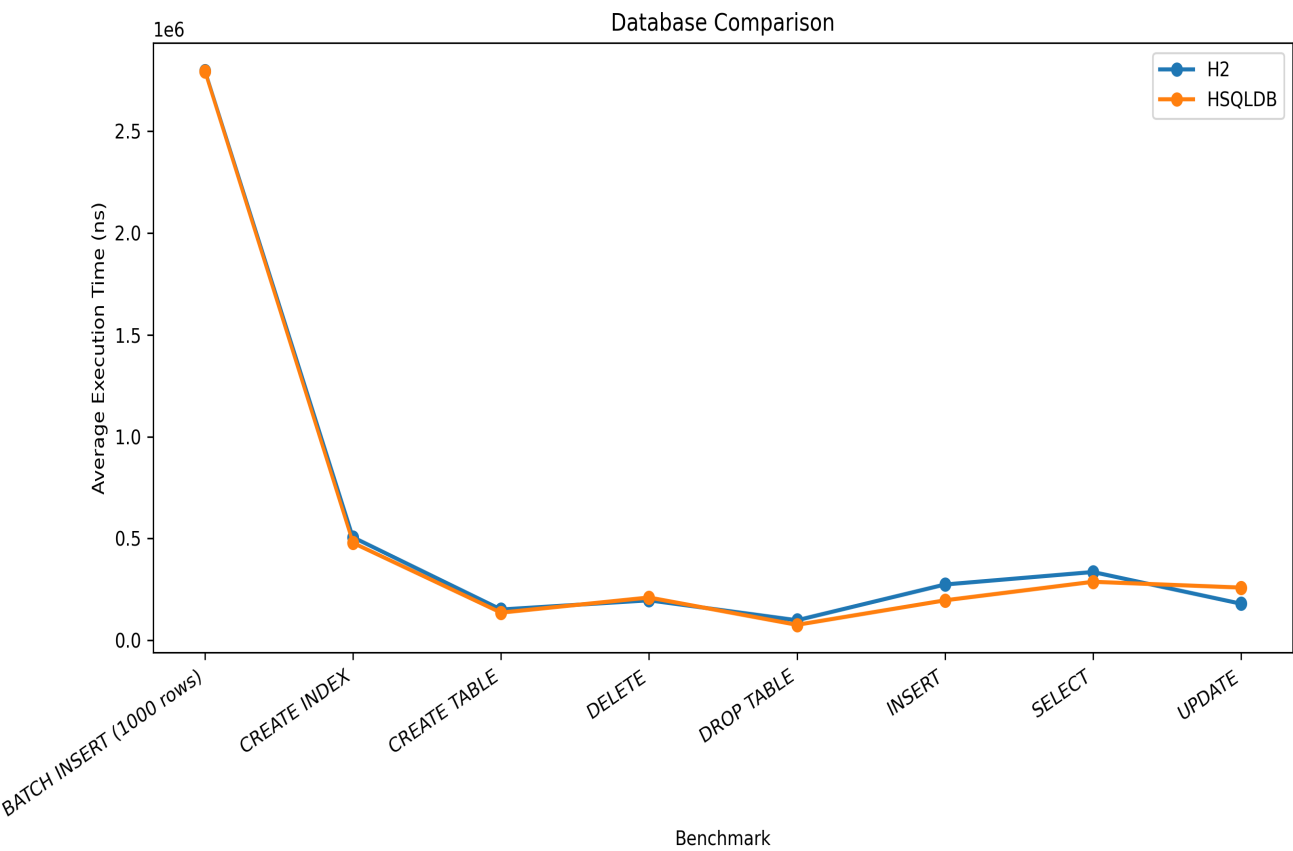
Standard Deviation



Variance



Database Comparison



Benchmark Winners

Benchmark	Winner	Average (ns)
BATCH INSERT (1000 rows)	HSQLDB	2,792,928
CREATE INDEX	HSQLDB	478,587
CREATE TABLE	HSQLDB	136,250
DELETE	H2	196,740
DROP TABLE	HSQLDB	76,289
INSERT	HSQLDB	196,635
SELECT	HSQLDB	287,690
UPDATE	H2	179,945

Overall Benchmark Score

Database	Benchmarks Won
HSQLDB	6
H2	2

Student's t-test

Benchmark	t-statistic	p-value	Significant?
BATCH INSERT (1000 rows)	0.011	0.991544	NO
CREATE INDEX	1.278	0.201448	NO
CREATE TABLE	2.988	0.002823	YES
DELETE	-2.430	0.015141	YES
DROP TABLE	6.084	0.000000	YES
INSERT	6.768	0.000000	YES
SELECT	2.986	0.002857	YES
UPDATE	-11.683	0.000000	YES

Interpretation

A p-value below 0.05 indicates that the observed performance difference between H2 and HSQLDB is statistically significant.

Automatic Conclusion

Summary

The benchmark suite executed 8 benchmark categories on 2 embedded database systems.

Based on the average execution time, **HSQldb** achieved the best overall performance, winning 6 benchmark categories.

The fastest benchmark measured was **DROP TABLE** executed on **HSQldb** with an average execution time of **76,289 ns**.

The slowest benchmark measured was **BATCH INSERT (1000 rows)** executed on **H2** with an average execution time of **2,796,589 ns**.

Overall, the benchmark results indicate that execution times remained stable after warm-up iterations while demonstrating measurable performance differences between the tested database systems.

This report was generated automatically from the benchmark CSV produced by the Java benchmark framework.